

# DELIVERY-AWARE SPARSE CAUSAL ROUTING FOR DISTRIBUTED MULTI-OBJECT TRACKING

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## ABSTRACT

We study sparse communication for distributed labeled multi-Bernoulli tracking when mobile formations change their physical neighbors. A causal routing policy retains a feasible formation tree, repairs it only when necessary, and combines local directed cycles with bidirectional gateways. The scheduled backbone uses  $N+2(F-1)$  posterior messages for  $N$  sensors in  $F$  formations. We distinguish this planned graph from packet delivery and label-specific fusion, and derive a finite-round sensitivity bound for exact geometric fusion. Paired 160-step development experiments with 24 and 36 sensors reduce attempted posterior bytes by 10.0% and 21.8% relative to fixed routing. Conditional localization RMSE decreases substantially, but set-error gains are smaller and a consistency tradeoff remains at 36 sensors. These results motivate delivery-aware receiver weighting while identifying the limits of connectivity alone as a design objective.

Index Terms— Multi-object tracking, labeled multi-Bernoulli, distributed fusion, dynamic topology, packet loss

## 1. INTRODUCTION

Mobile sensor formations exchange local posteriors to track objects beyond individual fields of view. As formations move, an initial route can lose an inter-formation connection even when an alternative physical link exists. Retaining many inputs improves redundancy but repeatedly transmits multi-object densities. The design problem is which posteriors reach which receivers within a finite tracking interval.

The labeled multi-Bernoulli (LMB) filter represents uncertain object existence and labeled spatial states [1]. Consensus LMB filtering uses Kullback–Leibler averaging (KLA), equivalently normalized geometric fusion, to combine these densities [2, 3]. Its classical equal-weight consensus limit assumes an appropriate stochastic matrix and repeated fusion of fixed inputs. A mobile tracker instead alternates local Bayesian updates with a limited

number of communication rounds. A connected scheduled graph consequently does not establish either successful delivery or an improvement in tracking error.

Randomized gossip relates averaging time to the spectrum of an appropriate stochastic matrix [4]. Event-triggered Bayesian filtering compares a posterior with its prediction from the last transmission to trigger communication [5]. Our focus is sparse route repair under changing physical feasibility and the inputs actually received for multi-object fusion, rather than a mixing-rate objective or a posterior-divergence trigger.

Different fields of view introduce another distinction: absence of a label at one sensor is not always evidence that the object has disappeared. Multi-view GCI methods address this mismatch at the density-fusion level [6]. Our focus is the communication structure and its receiver-side realization. The topology comparison keeps the underlying mixture-aware fusion approximation fixed. The routing policy uses physical link information rather than posterior-dependent learned fusion weights.

This paper develops a sparse causal backbone and analyzes how packet loss changes its intended fusion. The contributions of the present study are:

- a formation-level repair rule with an exact message count under a local-cycle and bidirectional-tree architecture;
- a separation of scheduled, packet-delivered and label-active graphs, with a finite-round perturbation bound for exact geometric fusion;
- paired fixed, full-repair and sparse-repair experiments exposing both the communication gains and the remaining set-estimation tradeoffs.

The results are development evidence from one seed at each scale. They motivate a controlled receiver-weighting experiment rather than a claim of statistically established cross-scenario superiority.

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## 2. SPARSE CAUSAL LMB ROUTING

### 2.1. Fusion and communication model

Sensor  $i$  maintains an LMB posterior  $\pi_i = \{(r_i^\ell, p_i^\ell)\}_{\ell \in \mathbb{L}_i}$ , where  $r_i^\ell$  is the existence probability and  $p_i^\ell(x)$  is a normalized spatial density. For a common active label and weights  $w_j \geq 0$ ,  $\sum_j w_j = 1$ , exact KLA satisfies [3]

$$\eta^\ell = \int \prod_j p_j^\ell(x)^{w_j} dx, \quad \bar{p}^\ell(x) = \frac{\prod_j p_j^\ell(x)^{w_j}}{\eta^\ell}, \quad (1)$$

$$\text{logit } \bar{r}^\ell = \sum_j w_j \text{logit } r_j^\ell + \log \eta^\ell. \quad (2)$$

Thus changing a neighbor affects both spatial overlap and existence. More mixing need not preserve the estimated target set.

Let  $G_t^{\text{phys}}$  describe current physical links, and let  $A_t$  be the scheduled directed graph, with rows denoting receivers and columns senders. The controller observes current positions, link reliabilities and its previous route, but no target truth or future trajectory. The implemented controller uses network-wide current geometry; decentralized acquisition and dissemination of this control information are outside the measured posterior-payload budget. Fusion itself is performed at each sensor.

### 2.2. Causal tree repair and sparse inputs

Sensors are partitioned into  $F$  formations. A candidate edge in the formation graph is usable when current physical sensor pairs can realize the required gateways. The policy first tries the preceding formation tree. When that tree or its gateway assignment is infeasible, it selects a current feasible tree lexicographically: minimize replaced formation edges, maximize bottleneck reliability, maximize total log reliability, then minimize distance. Gateway feasibility includes the receiver input constraints. This preserves feasible routes without using future link availability.

The sparse policy retains one directed cycle inside each formation. Each undirected edge of the formation tree is realized by one directed gateway in each direction; the two directions need not use the same sensor pair. Every sensor receives one dominant local input, and gateway receivers also receive one cross-formation input. The planned weights are 0.70 for the dominant input and 0.05 for a retained residual input, with the remaining mass assigned to self. The full-repair reference instead retains two nonself inputs per receiver. Removing its unused local residual input increases the sparse receiver's self weight by 0.05.

Proposition 1 (scheduled backbone). If every local cycle and every directed gateway is physically feasible, the

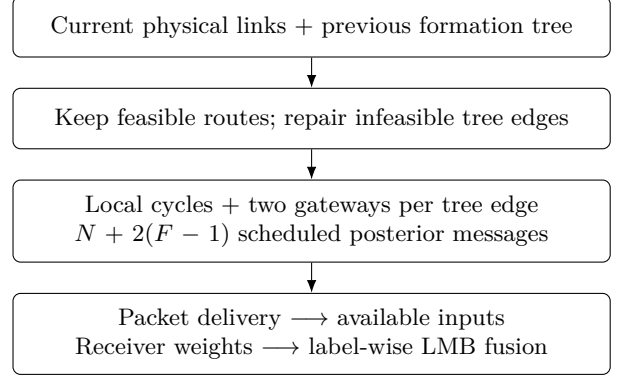


Fig. 1. The routing and fusion pipeline. Scheduled connectivity is a property of the third stage; actual information use also depends on delivery and label availability in the fourth stage.

scheduled sensor graph is strongly connected and contains exactly

$$M = N + 2(F - 1) \quad (3)$$

directed messages per round. Proof. The local cycles contribute  $N$  edges and provide directed paths between sensors in each formation. The tree contributes  $F - 1$  formation edges, each realized in both directions. Concatenating local paths and gateways gives a directed path between any two sensors. The edge sets are disjoint, giving (3). This is a count for the chosen architecture, not a global lower bound over all strongly connected directed graphs or a guarantee about delivered links.

## 3. PACKET DELIVERY AND FINITE-ROUND FUSION

### 3.1. Where a missing weight goes

Let  $W$  be a planned row-stochastic matrix with positive diagonal. For receiver  $i$ , let  $D_{ij} \in \{0, 1\}$  indicate delivery, set  $D_{ii} = 1$ , and define missing weight mass  $m_i = \sum_{j: D_{ij}=0} W_{ij}$ . The current reference implementation omits an unavailable input and uses

$$R_{ij} = \frac{W_{ij} D_{ij}}{1 - m_i}. \quad (4)$$

This differs from the planned removal of a local edge. An existing alternative receiver rule returns missing mass to self:

$$S_{ij} = W_{ij} D_{ij} + m_i \mathbf{1}\{i = j\}. \quad (5)$$

Both are row stochastic, and both satisfy  $\|R_{i:} - W_{i:}\|_1 = \|S_{i:} - W_{i:}\|_1 = 2m_i$ . Their distinction is therefore not a smaller row-wise perturbation norm. Rule (5) preserves the planned weight of each surviving neighbor. If a 0.70-weight dominant input is missing, a surviving

0.05-weight input becomes  $0.05/0.30 = 0.1667$  under (4), but stays at 0.05 under (5). The latter avoids amplification but can slow useful information transfer.

Neither rule preserves double stochasticity under arbitrary directed losses. For example, a symmetric two-node matrix with off-diagonal weight  $a$  becomes  $\begin{bmatrix} 1 & 0 \\ a & 1-a \end{bmatrix}$  after one direction fails under self fallback. Its column sums are  $1+a$  and  $1-a$ . Reciprocal scheduled gateways alone consequently do not establish equal-weight KLA consensus after packet loss.

### 3.2. A finite-round sensitivity bound

The following bound concerns exact geometric fusion of fixed positive densities on a common support. It does not assume new observations between consensus rounds. Write

$$q_\alpha(z) = \frac{\prod_j q_j(z)^{\alpha_j}}{\int \prod_j q_j(u)^{\alpha_j} du}, \quad \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1. \quad (6)$$

Assume the log-density oscillation on this support is bounded:  $\text{osc}(\log q_j) = \sup_z \log q_j(z) - \inf_z \log q_j(z) \leq L$ . For any two coefficient vectors  $\alpha, \beta$ ,

$$D_{\text{KL}}(q_\alpha \| q_\beta) \leq L \|\alpha - \beta\|_1. \quad (7)$$

To see this, put  $h = \sum_j (\alpha_j - \beta_j) \log q_j$ . Its oscillation is at most  $L \|\alpha - \beta\|_1$ . The log normalization ratio lies between  $\inf h$  and  $\sup h$ , so the log density ratio, and its expectation, are bounded above by the same oscillation.

Let  $P = \widetilde{W}_K \cdots \widetilde{W}_1$  and  $Q = W_K \cdots W_1$  be delivered and planned products. The induced infinity norm is sub-multiplicative and equals one for each stochastic factor. A telescoping expansion therefore gives

$$\max_i \|P_i - Q_i\|_1 \leq \sum_{k=1}^K \|\widetilde{W}_k - W_k\|_\infty \leq 2 \sum_{k=1}^K \max_i m_{i,k}. \quad (8)$$

Equations (7)–(8) connect lost weight mass to deviation from the scheduled geometric pool. They do not favor one fallback rule, imply lower ground-truth error, or establish uniform averaging. Gaussian densities on unbounded support need not satisfy the oscillation assumption. Mixture truncation, changing label support and intervening Bayesian updates additionally prevent treating this bound as a guarantee for the full implemented tracker.

## 4. EXPERIMENTS AND REMAINING TRADEOFFS

### 4.1. Paired protocol

We use a temporally coupled formation-braid simulation with four or six formations of six sensors, denoted M24

and X36. The scenes contain 16 and 24 targets over 160 time steps. Sensors within a formation share a heading, with a  $120^\circ$  field of view and 300 m sensing range. Detection probability is 0.88, clutter rate is 4 and measurement noise standard deviation is 7 m. Target handovers overlap periods of physical route failure. Each scale uses seed 1301; truth, measurements, directed delivery uniforms and filter randomness are paired across its arms.

The fixed baseline keeps its initial formation-tree edges; infeasible connections are dropped. Full causal repair retains two inputs per receiver and changes the formation tree when required. Sparse causal repair uses (3). These three arms share packet renormalization (4). A separate full-episode X36 run of (5) is in progress and is not included as a completed result.

All arms use a componentwise powered-Gaussian-mixture approximation to KLA: at most three input components, eight fused components and 256 component tuples. This preserves multiple modes but is not an exact arbitrary-GM power. Label-absence handling is held fixed across arms. The implementation uses one communication/fusion round per tracking step.

We report position-only Euclidean OSPA (E-OSPA), with order 2 and cutoff 150 m [7], and conditional Hungarian-matched position RMSE. E-OSPA is averaged over sensor–time pairs; RMSE averages their finite matched-set RMSE values, excluding a nonempty set paired with an empty set. The latter must be interpreted alongside absolute cardinality error. Focus consistency is mean pairwise E-OSPA between one fixed representative per formation over  $t = 40\text{--}140$ , not an all-node agreement measure. Communication cost is attempted posterior-payload bytes, including failed transmissions, but excluding routing-control traffic.

### 4.2. What the comparisons establish

Table 1 separates repair from sparsification. Full causal repair improves the three reported error measures over fixed routing at both scales, but increases attempted bytes by 10.052% on M24 and 5.708% on X36. The sparse policy instead saves 10.041% and 21.830% of attempted bytes. On M24 its E-OSPA and conditional RMSE improve by 2.403% and 46.190%, and focus consistency improves by 1.449%. However, the worst formation-wise RMSE change is a 24.085% degradation.

On X36 sparse repair improves E-OSPA by only 0.368% despite a 47.655% conditional-RMSE reduction. Mean absolute cardinality error rises from 18.456 to 18.597, and focus consistency worsens by 0.776%. Consequently, the RMSE decrease cannot be interpreted as a comparable improvement in complete target-set recovery. The large cardinality errors also show that the global target-set task remains difficult under limited

Table 1. Complete paired development episodes, seed 1301. Lower values are better in all four metric columns. Error measures are in metres; bytes are decimal MB of attempted posterior payload. Messages are scheduled, before delivery. Each scale has its own fixed baseline; no confidence intervals can be inferred from a single seed.

| Scale | Routing policy       | E-OSPA  | Cond. RMSE | Focus consistency | Payload | Messages/step |
|-------|----------------------|---------|------------|-------------------|---------|---------------|
| M24   | Fixed formation tree | 125.478 | 22.640     | 133.599           | 40.769  | 46–48         |
|       | Full causal repair   | 122.380 | 14.081     | 131.913           | 44.867  | 48            |
|       | Sparse causal repair | 122.462 | 12.183     | 131.664           | 36.676  | 30            |
| X36   | Fixed formation tree | 132.680 | 36.925     | 139.407           | 76.871  | 66–72         |
|       | Full causal repair   | 131.795 | 19.586     | 139.004           | 81.259  | 72            |
|       | Sparse causal repair | 132.192 | 19.329     | 140.489           | 60.090  | 46            |

views and finite communication.

A packet-only replay using the same delivery uniforms finds that the sparse graph is strongly connected after delivery on 41/160 M24 steps and 18/160 X36 steps. Planned matrices are non-doubly-stochastic on 17 and 22 steps, whereas packet-renormalized matrices are non-double on 122 and 143 steps. This replay assumes that a delivered packet supplies an input; empty payloads and label-specific censoring can further restrict fusion. The counts do not prove absence of connectivity over longer windows or identify the cause of the tracking differences.

#### 4.3. Limits and next discriminating experiment

The next controlled comparison retains the sparse topology and changes only the missing-input rule to (5). It asks whether avoiding surviving-neighbor amplification improves E-OSPA, cardinality and consistency while retaining the byte advantage. Its outcome is not assumed by the analysis. A positive result must be checked on M24 before further tuning. Frozen multi-seed tests and a separate non-radial zipper-merge scene are then required to evaluate reproducibility and geometry dependence.

The current evidence does not establish statistical significance, end-to-end control-inclusive bandwidth savings, or a decentralized routing implementation. The message count and perturbation arguments are analytical properties of the stated construction and assumptions; their usefulness for set estimation must be demonstrated rather than inferred from connectivity.

## 5. CONCLUSION

Sparse causal routing provides a concrete communication advantage over a fixed formation tree in two paired development cases. Its joint tracking benefit remains incomplete: conditional localization gains coexist with cardinality, consistency and formation-tail tradeoffs. The resulting research direction is delivery-aware sparse fusion, with actual received inputs as the link between topology design and finite-round LMB estimation.

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## 6. REFERENCES

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